Perpendicular Lines — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: Perpendicular lines*

Key Idea

Perpendicular lines meet at a **right angle** ($90°$). Symbol: a **$\perp$** b.

In a picture, a little **square** marks a right angle.

☞ *Video: Linear pair of congruent angles*

Watch Me — Worked Example

$\angle 1$ and $\angle 2$ form a linear pair: $m\angle 1 + m\angle 2 =$ **$180°$**

If $\angle 1 \cong \angle 2$, each is $180 \div 2 =$ **$90°$**.

So two lines that form a linear pair of congruent angles are **perpendicular**.

☞ *Video: Two lines perpendicular to the same line*

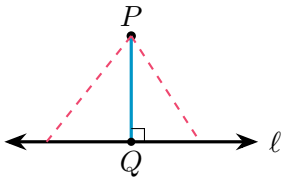
Key Idea

In a plane: if $a \perp t$ and $b \perp t$, then a **$\parallel$** b.

Why: both make $90°$ **corresponding** angles with t, so the angles are congruent.

☞ *Video: Distance from a point to a line*

Key Idea



Drop a segment from P straight to the line: $\overline{PQ}$ **$\perp$** ℓ.

Every slanted path is **longer** than PQ.

Distance from a point to a line = length of the **perpendicular** segment — the **shortest** path.

☞ *Video: Watch out*

Common Mistake

Only the **perpendicular** segment measures the distance to a line — not a slanted one.

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☞ *Video: Your turn*

Your Turn

Point $P(3,5)$ and the line $y=1$. What is the distance from P to the line?

Go straight down to $(3,1)$. Distance $= 5-1 =$ 4

Notice

Remember: Perpendicular lines meet at $90°$. A linear pair of congruent angles means perpendicular. Distance to a line $=$ the perpendicular segment, the shortest path.

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